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Original Research

Real-World Utilization Patterns of Immune Checkpoint Inhibitors Based on the National Health Insurance Service Data in Korea

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ABSTRACT

Purpose: This study was conducted to comprehensively understand the utilization patterns of immune checkpoint inhibitors (ICIs) in Korea, including their use in combination with traditional anticancer therapies.
Methods: We investigated the utilization of ICIs using claims data from Korea between 2017 and 2022. Patients with cancer were included in the study if they received at least one dose of ICIs, defined by drug codes in the claims data. We used descriptive statistics to identify patterns of ICIs use in combination with other anticancer therapies and ICIs use by year and cancer type.
Findings: During the study period, 41,208 patients received at least one dose of ICIs. Lung cancer (66.6%), urinary tract cancer (11.3%), and liver/biliary tract cancer (10.6%) were the most frequent cancer types. The prescription of ICIs particularly surged between 2020 and 2022 for liver/biliary tract cancer. Between 2017 and 2022, the use of ICIs increased from 1,155 to 15,034, with an increase every year. Since 2020, the use of ICIs in combination with other anticancer therapies and concurrent use of ICIs has increased sharply.
Implications: The number of patients utilizing ICIs in Korea has been steadily increasing. As regulatory approval of indications for ICIs has expanded, so has the range of indications for ICIs. In addition, the combination of ICIs with other anticancer therapies and the concurrent use of ICIs has increased in recent years, reflecting expanded treatment options for advanced cancers.

Introduction

Cancer is a major global health issue and remains the leading cause of death worldwide, accounting for approximately 10 million deaths in 2022 alone.[1,2] While conventional chemotherapy has improved survival for many patients, it is often associated with systemic toxicity, limited efficacy, and anticancer drug resistance, particularly in those with advanced-stage cancer.

Immune checkpoint inhibitors (ICIs) have transformed the landscape of cancer treatment by offering new options for patients with advanced cancers by enhancing the immune system's ability to recognize and attack tumor cells.[3] Since the approval of ipilimumab, a cytotoxic T lymphocyte-associated protein-4 (CTLA-4) inhibitor, in 2011, programmed death-1 (PD-1) receptor inhibitors such as nivolumab, pembrolizumab, and cemiplimab; and programmed death ligand-1 (PD-L1) inhibitors such as atezolizumab, avelumab, and durvalumab have been introduced. These agents have broadened therapeutic options for various cancers, including non–small cell lung cancer (NSCLC), metastatic melanoma, renal cancer, and other solid and hematological malignancies.[4,5] ICIs can be administered as

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monotherapy, in combination with other ICIs (as combined PD-1 and CTLA-4 inhibitors; nivolumab plus ipilimumab), or in combination with chemotherapy or radiation therapy.[6,7]

As the indications for ICIs continue to expand, their use in clinical practice is expected to increase accordingly. However, previous studies on ICI utilization have focused mainly on specific cancer types,[8-11] and comprehensive nationwide real-world utilization patterns, including evolving indications and treatment regimen, remain limited. A prior U.S.-based study using claims data from 2008 to 2019 focused primarily on immune-related adverse events and did not examine utilization patterns, particularly concerning combination therapies.[12] Moreover, treatment patterns may vary by countries due to differences in cancer epidemiology. For instance, melanoma is a relatively common indication for ICIs in the US, but is much less prevalent in Korea.[2,9] Healthcare system factors such as drug approval and reimbursement policies may also influence ICI utilization. In Korea, reimbursement for ICIs was initiated in August 2017 as second-line treatment for NSCLC and urothelial carcinoma, and as first-line treatment for malignant melanoma. Since then, coverage has expanded to include additional cancer types such as gastric, esophageal, colorectal, endometrial, and head and neck cancers. In addition, off-label use may be approved on a case-by-case basis through evaluation by the national health authority.

This study aimed to comprehensively examine the utilization of all types of ICIs in Korea between 2017 and 2022 using claims data from the National Health Insurance Service (NHIS). Specifically, we assessed utilization patterns of ICIs monotherapy and combination therapies with other anticancer therapies, trends by cancer type and year, and overall survival (OS) among ICIs users.

Methods

Data Source and Study Population

In Korea, the mandatory health insurance system is in place for all Korean residents, with 97% of the population enrolled. We used a customized database from 2017 to 2022 provided by the NHIS in Korea.[13] This database included all individuals who were diagnosed with cancer according to the 10th edition of the International Classification of Diseases (ICD-10; C00-C97, D00-D09, D32-D33, D37-D48, D76.0) codes and the rare and intractable disease (RID) codes (V193 or V194) indicating confirmed cancer diagnosis. This study population comprised those who had at least one claim for ICIs use. The ICIs are coded using World Health Organization (WHO) Anatomical Therapeutic Chemical (ATC) codes in the claims data as follows: atezolizumab (ATC code: L01FF05), avelumab (L01FF04), durvalumab (L01FF03), ipilimumab (L01FX04), nivolumab (L01FF01), and pembrolizumab (L01FF02).

Study Variables

Anticancer Therapies

To identify other anticancer therapies used with ICIs, we defined chemotherapy and targeted therapy using WHO ATC codes and radiotherapy using reimbursement procedure codes from the Korean Health Insurance Review and Assessment Service (Table S1).

ICI Indications, Patient Characteristics, Comorbidities, and Overall Survival (OS)

Cancer types were defined based on the date of the first prescription of the ICIs, using ICD-10 codes as follows: skin cancer (C43-44), lung cancer (C34), mesothelioma of pleura (C45.0), urinary tract cancer (C64-C68), colorectal cancer (C18-C20), breast cancer (C50), cervical cancer (C53), liver/biliary tract cancer (C22-C24), head and neck cancer (C00-C14, C30-C32), Hodgkin lymphoma (C81), and endometrial cancer (C54).

Patient characteristics included age, sex, income level, Charlson comorbidity index (CCI), drug utilization patterns of ICIs, and cancer type. The individual income level was divided into 4 quartiles (Q1-Q4). Data regarding the history of anticancer therapy and CCI were assessed within 1 year before the index date, and the CCI score was calculated excluding cancer-related components. In addition, all-cause mortality and date of death were used to estimate OS.

Statistical Analysis

The demographic characteristics of the study population during the study period are presented using descriptive statistics. Categorical variables are presented as the number (percentage), and continuous variables are presented as the mean (standard deviation [SD]). We analyzed the frequency of use of ICIs in combination with other anticancer therapies, and the use of ICIs by year and cancer type. Median survival was estimated from first date of ICI treatment to all-cause death or end of database (December 31, 2022). Additionally, Kaplan-Meier curves were plotted to evaluate OS, and differences in survival were assessed using the log-rank test according to treatment type (ICI monotherapy vs. ICI in combination with other anticancer therapies), number of ICI administrations (single or multiple doses), and cancer type. Statistical analyses were performed using SAS Enterprise software version 7.1 (SAS Inc., Cary, North Carolina) and R Studio software version 4.3.1 (R Studio Inc., Boston, Massachusetts).

Results

Study Population and Demographic Characteristics

During the study period, 982,355 patients had a confirmed diagnosis of cancer in Korea. Among them, we identified 41,216 patients who received at least one dose of ICIs. After excluding patients with missing age or sex information ($n = 8$), we analyzed 41,208 patients.

Their mean age was 68.0 (±10.3) years, and the proportion of males (76.3%) was higher than that of females. The proportion of patients with a CCI score of ≥ 2 was 71.2%, and the mean CCI score was 2.8 (±2.1) (Table 1).

Indications for ICIs Use

The most frequent cancer types among the study population were lung cancer (66.6%), urinary tract cancer (11.3%), and liver/biliary tract cancer (10.6%). The number of ICIs users for liver/biliary tract cancer sharply increased from 334 in 2020 to 2,755 in 2022, and the number of users for urinary tract cancer showed a steady annual increase. By 2022, the number of cases in which ICIs were used in combination with other anticancer therapies steadily increased for patients with lung cancer, surpassing the number of cases in which ICIs were not combined with other anticancer therapies. Combination ICIs therapy also sharply rose for patients with liver/biliary tract cancer since 2021 (Table 2, Figure 1). Table S2 presents the detailed annual frequencies of ICI use by indication.

Patterns of ICIs Utilization by Drug and Year

Among patients who started ICIs, 68.9% ($n = 28{,}387$) used ICIs without other anticancer therapies and 31.1% ($n = 12{,}821$) used ICIs in combination with other anticancer therapies. The most frequently used ICIs were atezolizumab (41.9%), followed by pembrolizumab (34.2%) and nivolumab (16.8%) (Table 3).

Among ICIs used without any other class of anticancer therapy, the most frequently used ICIs were atezolizumab (35.3%), pembrolizumab (32.4%), and nivolumab (22.2%). When ICIs were combined with other anticancer therapies, the most frequently used ICIs were atezolizumab (56.5%) and pembrolizumab (38.2%), mostly combined with chemotherapy (18.4%) (Table 3, Figure 2). Atezolizumab was most often used in lung and liver/biliary tract cancer and pembrolizumab in

Table 1
Baseline characteristics of users of immune checkpoint inhibitors.

Characteristics	N (%)
Sociodemographic information	
Sex	
Male	314,23 (76.3)
Age, mean (SD)	68.0 (10.3)
≤ 39 years	370 (0.9)
40–49 years	1,581 (3.8)
50–59 years	5,860 (14.2)
60–69 years	13,970 (33.9)
70–79 years	14,209 (34.5)
≥ 80 years	5,218 (12.7)
Income level	
0–5	10,109 (24.5)
6–10	7,270 (17.6)
11–15	9,726 (23.6)
16–20	13,939 (33.8)
Missing	164 (0.4)
Medical aid	2,579 (6.3)
Geographic region	
Metropolitan	10,295 (25.0)
Seoul	7,397 (18.0)
Other	23,516 (57.1)
CCI score without contribution from cancer (1 year before treatment), mean (SD)	2.8 (2.1)
0–1	11,883 (28.8)
≥ 2	29,325 (71.2)
Cancer type	
Skin cancer	2,069 (5.0)
Lung cancer	27,453 (66.6)
Mesothelioma of pleura	15 (0.0)
Urinary tract cancer	4,646 (11.3)
Colorectal cancer	250 (0.6)
Esophageal cancer	286 (0.7)
Stomach cancer	1,014 (2.5)
Breast cancer	190 (0.5)
Cervical cancer	220 (0.5)
Liver/biliary tract cancer	4,370 (10.6)
Head and neck cancer	714 (1.7)
Hodgkin lymphoma	41 (0.1)
Endometrial cancer	164 (0.4)

Table 3
Drug utilization patterns of immune checkpoint inhibitors and history of anticancer therapy.

Variable	N (%)
History of anticancer therapy	
Anticancer therapy before ICIs use (Within 1 year before the first ICIs use)	
Chemotherapy	14,178 (34.4)
Radiotherapy	1,620 (3.9)
Targeted therapy	715 (1.7)
Chemotherapy + Radiotherapy	8,401 (20.4)
Chemotherapy + Targeted therapy	1,968 (4.8)
Radiotherapy + Targeted therapy	265 (0.6)
Chemotherapy + Radiotherapy + Targeted therapy	860 (2.1)
Drug utilization patterns of ICIs	
ICIs	
Atezolizumab	17,268 (41.9)
Avelumab	20 (0.0)
Durvalumab	1,992 (4.8)
Ipilimumab	2 (0.0)
Nivolumab	6,908 (16.8)
Pembrolizumab	14,103 (34.2)
Ipilimumab + Nivolumab	915 (2.2)
Types of anticancer therapies used in combination with ICIs	
Chemotherapy	7,569 (18.4)
Radiotherapy	1,420 (3.4)
Targeted therapy	2,939 (7.1)
Chemotherapy + Radiotherapy	755 (1.8)
Chemotherapy +Targeted therapy	17 (0.0)
Radiotherapy + Targeted therapy	115 (0.3)
Chemotherapy + Radiotherapy+ Targeted therapy	6 (0.0)
No. of ICIs cycles, mean (SD)	
Overall	7.3 (8.7)
2017	3.2 (2.1)
2018	6.0 (5.7)
2019	5.6 (5.6)
2020	5.4 (5.2)
2021	5.6 (5.2)
2022	5.9 (5.1)
Dosing interval of ICIs (for patients who were prescribed ICIs more than twice), mean (SD)	22.3 (10.3)

N, number of patients.
Abbreviation: ICIs, immune checkpoint inhibitors; SD, standard deviation.

skin cancer. Nivolumab plus ipilimumab was mostly used in urinary tract cancer (Figure 3).

Between 2017 and 2022, the number of patients using ICIs increased from 1,155 to 15,034, with the number increasing each year. The use of ICIs without other anticancer therapies increased until 2021 and then decreased in 2022, whereas the use of ICIs in combination with other anticancer therapies has been rising since 2020 (Figure 2A). In 2022, the use of atezolizumab decreased when ICIs were not used combined with other anticancer therapies (Figure 2B), with a sharp increase in the use of atezolizumab and pembrolizumab in combination with other anticancer therapies (Figure 2C). When ICIs were not combined with any other class of anticancer therapy, the use of ipilimumab in combination with nivolumab has been sharply increasing since 2020 (Figure 2B). Since 2020, the proportion of patients using ICIs in combination with chemotherapy has decreased, whereas the proportion of targeted therapy in combination with ICIs has increased (Figure S1).

Table 2
Type of cancer for which immune checkpoint inhibitors were used in each year.

Variable	2017 N (%)	2018 N (%)	2019 N (%)	2020 N (%)	2021 N (%)	2022 N (%)
Skin cancer [C43-44]	4 (0.3)	530 (15.4)	393 (8.1)	358 (5.1)	407 (4.2)	377 (2.5)
Lung cancer [C34]	1,132 (96.5)	2,410 (70.2)	3,518 (72.3)	5,027 (71.2)	6,049 (62.3)	9,317 (61.3)
Mesothelioma of pleura [C45.0]	0 (0.0)	0 (0.0)	0 (0.0)	4 (0.1)	6 (0.1)	5 (0.0)
Urinary tract cancer [C64-C68]	9 (0.8)	295 (8.6)	621 (12.8)	910 (12.9)	1,285 (13.2)	1,526 (10.0)
Colorectal cancer [C18-C20]	5 (0.4)	23 (0.7)	22 (0.5)	50 (0.7)	67 (0.7)	83 (0.5)
Esophageal cancer [C15]	1 (0.1)	13 (0.4)	28 (0.6)	49 (0.7)	81 (0.8)	114 (0.8)
Stomach cancer [C16]	6 (0.5)	52 (1.5)	91 (1.9)	144 (2.0)	366 (3.8)	355 (2.3)
Breast cancer [C50]	1 (0.1)	7 (0.2)	4 (0.1)	15 (0.2)	20 (0.2)	143 (0.9)
Cervical cancer [C53]	1 (0.1)	6 (0.2)	24 (0.5)	65 (0.9)	74 (0.8)	50 (0.3)
Liver/biliary tract cancer [C22-C24]	4 (0.3)	56 (1.6)	126 (2.6)	334 (4.7)	1,095 (11.3)	2,755 (18.1)
Head and neck cancer [C00-C14, C30-C32]	10 (0.9)	40 (1.2)	30 (0.6)	82 (1.2)	178 (1.8)	374 (2.5)
Hodgkin lymphoma [C81]	0 (0.0)	0 (0.0)	0 (0.0)	2 (0.0)	10 (0.1)	29 (0.2)
Endometrial cancer [C54]	0 (0.0)	1 (0.0)	6 (0.1)	25 (0.4)	70 (0.7)	62 (0.4)
Total N (%)	1,173 (100.0)	3,433 (100.0)	4,863 (100.0)	7,065 (100.0)	9,708 (100.0)	15,190 (100.0)

N, number of patients.

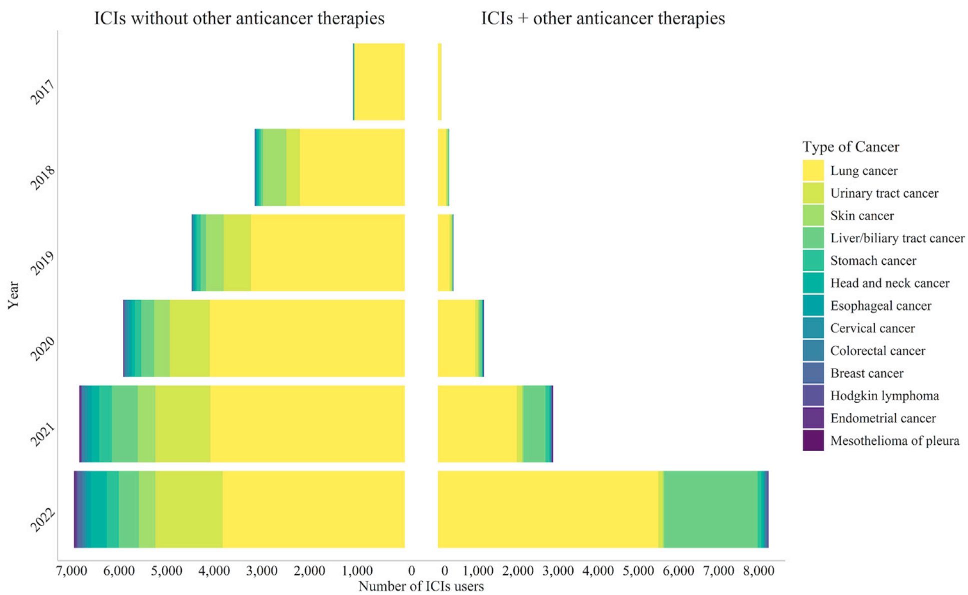

Figure 1. Annual distribution of cancer types treated with immune checkpoint inhibitors, classified by ICIs without other anticancer therapies and combination therapy. ICIs, immune checkpoint inhibitors.

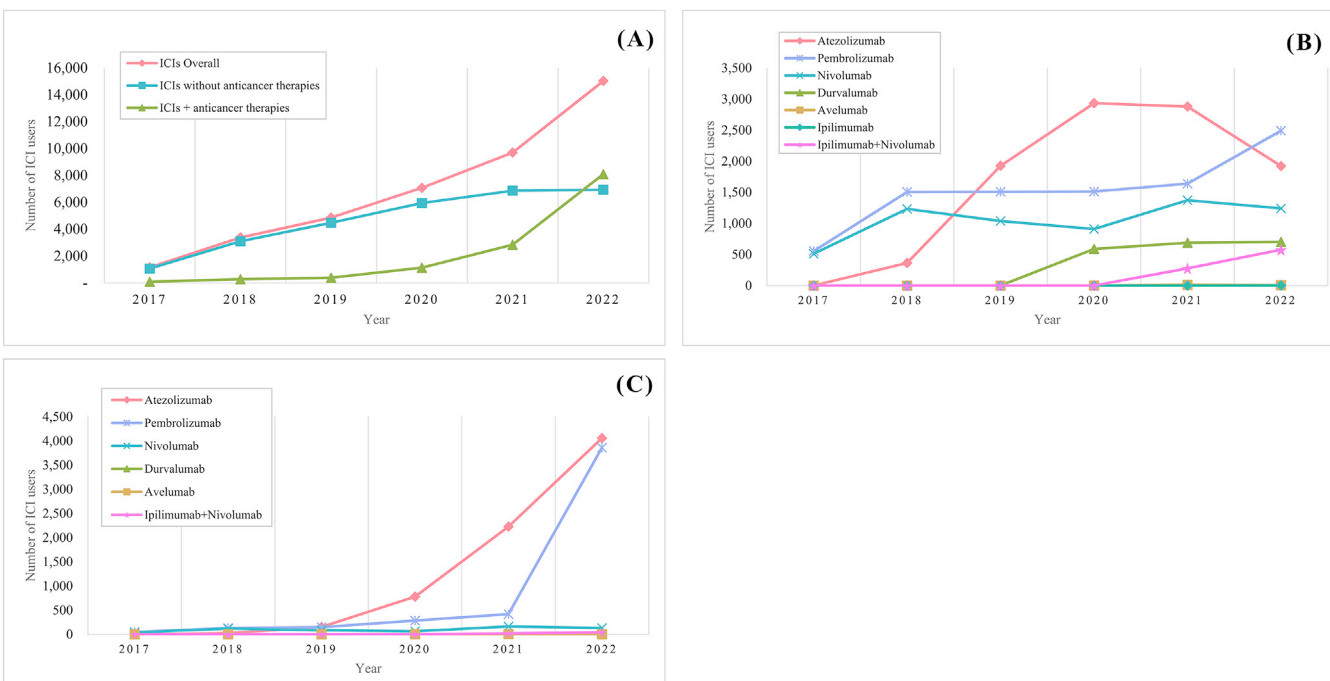

Figure 2. Immune checkpoint inhibitors usage by year. (A) Total ICIs (B) ICIs without other anticancer therapies (C) ICIs + other anticancer therapies. ICIs, Immune checkpoint inhibitors.

The most frequently used anticancer therapies before ICIs use were chemotherapy (50.6%), chemotherapy + radiotherapy (30.0%), and chemotherapy + targeted therapy (7.0%). Among Patients who had previously received chemotherapy, the most frequently used ICIs were atezolizumab, followed by pembrolizumab and nivolumab. Durvalumab was mostly used in patients who had previously received chemotherapy + radiotherapy (Figure S2).

Overall survival

The median OS was 301 days (95% CI: 295–307) for overall ICIs users (Figure 4A), 306 days (95% CI: 298–312) among those who received ICIs without other anticancer therapies, and 283 days (95% CI: 272–293) among those who received ICIs in combination with other anticancer therapies (log-rank $p < 0.05$) (Figure 4B). The median survival (OS) was 378 days (95% CI: 370–385) for patients who received multiple doses, and 44 days (95% CI: 43–45) for those who received single dose ICIs. By cancer type, OS was highest for skin cancer (median: 540 days; 95% CI: 500–588), followed by urinary tract cancer (median: 326 days; 95% CI: 307–352), lung cancer (median: 293 days; 95% CI: 286–300), and liver/biliary tract cancer (median: 274 days; 95% CI: 258–286) (log-rank $p < 0.05$) (Figure 4D)

Discussion

This study involved a comprehensive analysis of the utilization patterns of ICIs from 2017 to 2022, using claims data from the NHIS in Korea. The findings of this study indicate that the use of ICIs has steadily increased during the study period, and the use of ipilimumab in combination with nivolumab particularly has risen significantly in recent

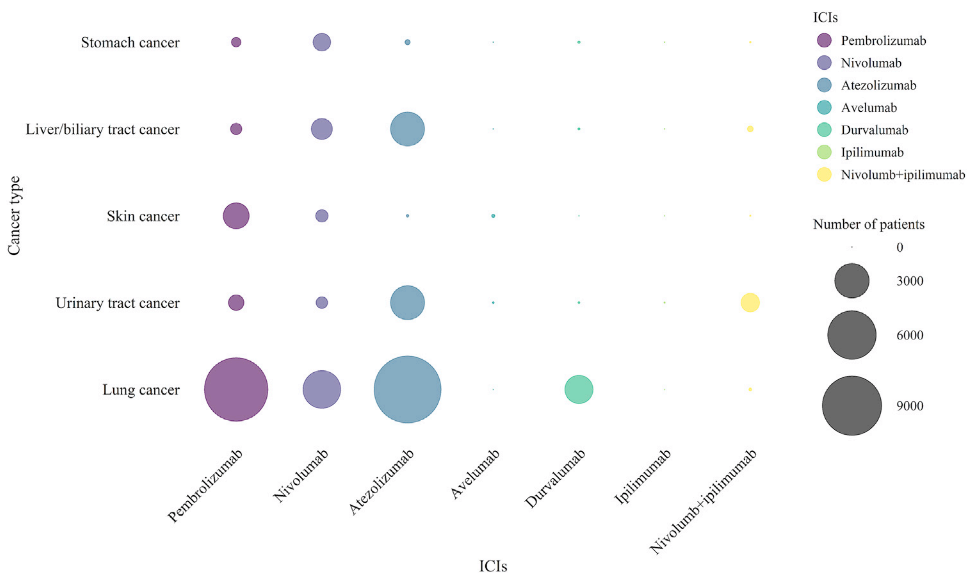

Figure 3. Patterns of immune checkpoint inhibitors usage for various cancer types. ICIs, immune checkpoint inhibitors.

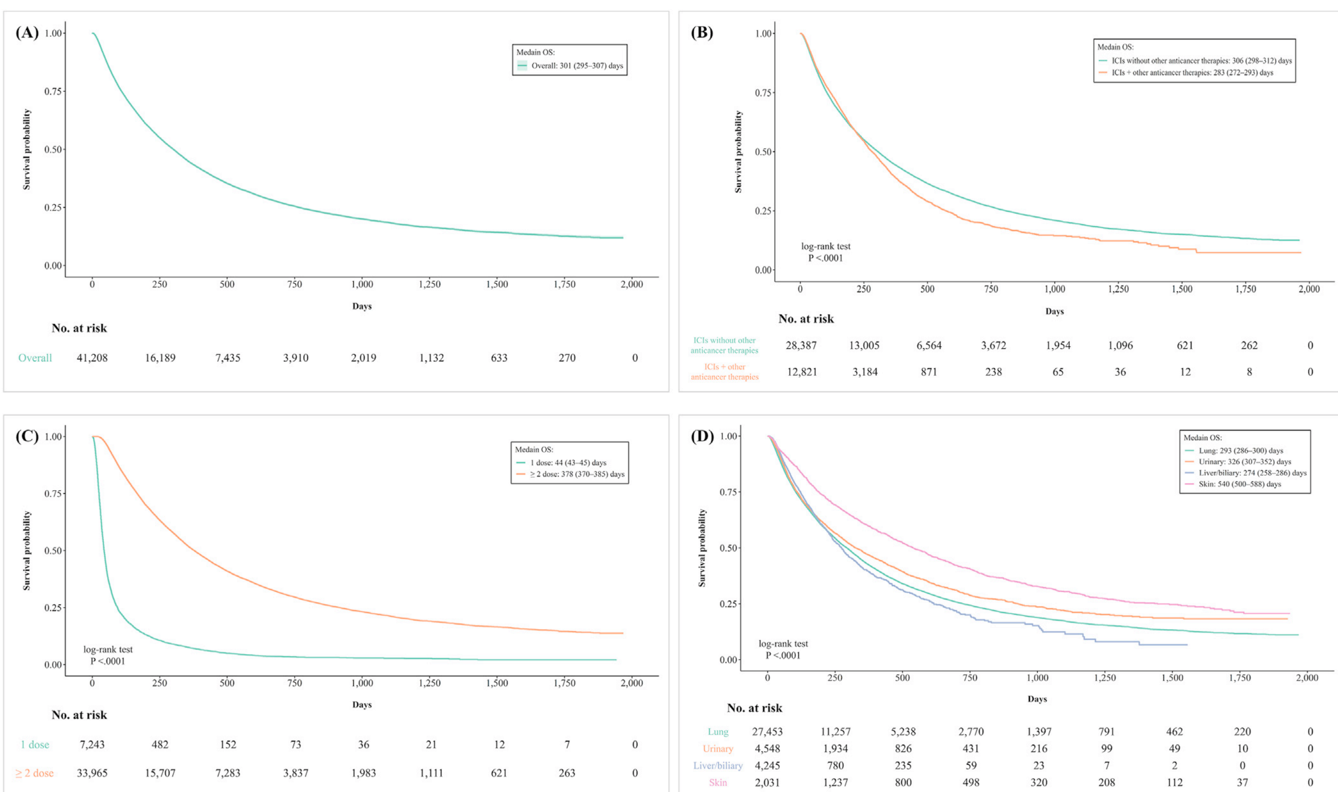

Figure 4. Kaplan-Meier analysis of overall survival in (A) overall ICI users, and according to (B) whether ICIs used combined with other anticancer therapies, (C) number of ICI doses, and (D) cancer types. ICIs, Immune checkpoint inhibitors; OS, overall survival.

years. The use of ICIs in treating cancers such as those of the urinary tract and liver/biliary tract has increased. This trend is supported by reimbursement coverage over drugs such as atezolizumab and pembrolizumab, particularly for the treatment of metastatic urothelial carcinoma for which standard platinum-based chemotherapy have shown poor efficacy[14–16]

The use of ICIs was higher in males (76.3%), particularly for the treatment of lung and urinary tract cancers, which is similar to the results of a Swiss study, in which male patients accounted for 68.7%.[17] In addition, lung cancer was the most common cancer type treated with ICIs in both studies. Atezolizumab, nivolumab, and pembrolizumab—which have been reimbursed since 2017—accounted for the majority of ICI use across cancer types. Durvalumab (approved in 2018 and reimbursed from 2020) and avelumab (approved in 2019 and reimbursed from 2020) showed a small but increasing trend in use. Although the number of cases was relatively low, the use of ICIs in breast, endometrial, esophageal cancers as well as Hogkins' lymphoma has been steadily increasing, likely reflecting off-label use. Our findings are consistent with global trends showing an increase in ICIs use. In this study, the number of patients using ICIs increased each year from 1,155 in 2017 to 15,034 in 2022 during the study period. A previous study on the US Medicaid program similarly reported an increase in ICIs use from

94 in 2011 to 462,049 in 2021, demonstrating the acceptance of ICIs as a foundation for cancer treatment.[18] A notable finding of this study was the decrease in the use of atezolizumab in 2022, alongside an increase in the use of pembrolizumab. This shift can be attributed to the substantial rise in the share of pembrolizumab use in lung and urinary tract cancers. In this regard, the reimbursement of pembrolizumab monotherapy and combination therapy for NSCLC in March 2022, and the reimbursement of pembrolizumab monotherapy for urothelial carcinoma in August 2022 in South Korea, are likely to have contributed to this trend.[19]

In this study, we identified a recent expansion in the use of ipilimumab in combination with nivolumab. This is supported by studies highlighting not only the significant increase in acceptance of combination therapy of ipilimumab and nivolumab for a range of cancer types but also its improved efficacy and safety compared to monotherapy.[7,20]

Recent studies show the growing trend toward using a combination of ICIs with other anticancer therapies to overcome resistance and improve treatment outcomes in various cancer types.[21,22] Preclinical studies showed that chemotherapeutic agents increase PD-L1 expression in tumor cells, making the combination of ICIs and chemotherapy effective.[23] Clinical trial evidence supporting the efficacy of pembrolizumab in combination with chemotherapy in patients with nonsquamous NSCLC led to the FDA approval of the first ICI in combination with chemotherapy in May 2017.[24] In May 2020, the combination of atezolizumab and bevacizumab was approved for the treatment of hepatocellular carcinoma,[25] which is consistent with the use trend for atezolizumab and targeted therapy in combination with ICIs and other anticancer therapies since 2020 in this study. The National Comprehensive Cancer Network guidelines also recently promoted the use of ICIs in combination with other anticancer therapies in liver/biliary tract cancer.[26,27]

In this study, the group receiving ICIs without other anticancer therapies had a higher median OS (306 days) compared to those receiving ICIs in combination with other anticancer therapies (283 days). One possible explanation for the relatively lower OS in the combination therapy group is that these patients may have had more advanced disease or higher clinical risk.[22,28] However, this result should be interpreted with caution. The imbalances in cancer types between the 2 groups may have influenced the observed survival outcomes. For example, the proportion of patients with liver/biliary tract cancer was substantially higher in the combination therapy group (23.1%) than in the monotherapy group (5.0%), which reflects recommendations of atezolizumab-bevacizumab in hepatocellular carcinoma. Conversely, urinary tract cancer was more prevalent in the monotherapy group (15.1%) compared to the combination therapy group (2.9%) (Table S2). Moreover, while PD-L1 expression level is a critical biomarker of ICI response, and combination therapy with chemotherapy is generally recommended for patients with lower PD-L1 expression, particularly in NSCLC,[29] this variable was not available in our database. Although some PD-1 inhibitors (nivolumab and pembrolizumab) are reimbursable only for PD-L1-positive patients in Korea, PD-L1 status could not be incorporated into the analysis, which may act as a confounder in our OS analysis.

Patients who received only a single dose of ICIs had significantly lower median OS (44 days) compared to those who received 2 or more doses (378 days). This may reflect underlying differences in disease severity or patient frailty, as supported by the higher mean CCI score observed in the single-dose group (mean CCI: 3.04) compared to those who received multiple doses (mean CCI: 2.82). Median OS by cancer type was highest in patients with skin cancer (540 days), followed by urinary tract cancer (326 days), and liver/biliary tract cancer (274 days). These findings are consistent with previous studies reporting median OS of 22.2 months for malignant melanoma, 12.2 months for urothelial carcinoma, and 11.1 months for non–small-cell lung cancer.[30] The particularly favorable OS in skin cancer may reflect the intrinsic tumor biology and its responsiveness to ICI therapy. However, given the differences in baseline characteristics among the comparison groups, and the lack of data on cancer staging and functional status, these survival results should not be interpreted as a direct measure of ICI treatment effect.

A key strength of this study is that it is the first study to comprehensively examine the utilization patterns of ICIs in Korea. We also used a comprehensive 6-year dataset from the NHIS, which covers approximately 97% of the Korean population. This large population-based database enables analysis of the real-world use of ICIs, providing valuable insights into cancer treatment trends in Korea.

Despite these strengths, some limitations must be acknowledged. As a retrospective analysis based on claims data, it is subject to potential inaccuracies in diagnostic coding and missing claims. However, in our data, the cancer diagnosis based on both the ICD-10 codes and RID cancer registration codes was relatively accurate, because these coding systems necessitate reporting of only confirmed cancer cases.[31] However, given the nature of a claims database, treatments beyond national insurance coverage, clinical information on cancer stage, PD-L1 expression level, exact doses of ICIs, treatment prognosis, and functional status were not considered in this study. In addition, while we were able to evaluate OS, progression-free survival could not be assessed, as disease progression is not reliably identifiable in the claims database.

Conclusion

In conclusion, this study comprehensively analyzed the real-world application of ICIs in Korea over 6 years by utilizing data from the NHIS. In this study, we found that the overall use of ICIs has steadily increased in Korea, and expanded regulatory approvals have led to expanded indications for cancers such as urinary tract and liver/biliary tract cancer. These findings suggest a new treatment modality for cancer patients. Furthermore, the use of ICIs in combination with other anticancer therapies has seen an upward trend in recent years. We also observed a sharp increase in the use of the combination of nivolumab and ipilimumab in recent years, reflecting the growing complexity of cancer treatment regimens. Understanding the utilization trends of ICIs from this study will guide future research and policy decisions in oncology.

Ethics Approval and Consent to Participate

The study protocol was approved by the Institutional Review Board of Chung-Ang University (IRB No. 1041078-20230411-HR-103). The need for informed consent was waived because this study analyzed anonymous claims data provided for research by the Korean National Health Insurance Service.

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Data Availability Statement

The data are available upon request as they contain privacy concerns of individuals in the claims database. Data can only be obtained upon permission from the National Health Insurance Service (https://nhiss.nhis.or.kr/bd/ay/bdaya001iv.do).

Declaration of competing interest

None.

CRediT authorship contribution statement

Eunji Kim: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yu-Seon Jung:** Writing – review & editing, Validation, Methodology, Investigation, Conceptualization. **Jongmin Lee:** Writing – review & editing. **Dal Ri Nam:** Writing – review & editing. **Seung-Hun You:** Writing – review & editing. **Ju Won Lee:** Writing – review & editing. **Sook Ryun Park:** Writing – review & editing, Validation, Investigation, Conceptualization. **Ji Seon Oh:** Writing – review & editing, Validation, Investigation, Conceptualization. **Ye-Jee Kim:** Writing – review & editing, Validation, Investigation, Conceptualization. **Eun-Jung Jo:** Writing – review & editing, Validation, Investigation, Conceptualization. **Nakyung Jeon:** Writing – review & editing, Validation, Investigation, Conceptualization. **Won-Jung Jung:** Writing – review & editing. **Sun-Young Jung:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.clinthera.2025.07.014.

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